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#liner search
import time
import matplotlib.pyplot as plt
def linear_search(arr,n,key):
    for i in range(n):
        if arr[i]==key:
            return i
    return -1

def linear_key(r):
    results=[]
    for _ in range(r):
        n=int(input("Enter the number of elements:"))
        arr=list(map(int,input("\nEnter the elements of an
array:").split()))
        key=int(input("\nEnter the key element to be searched:"))

        repeat=10000
        result=-1
        start=time.time()
        for _ in range(repeat):
            result=linear_search(arr,n,key)

        end=time.time()
        if result!=-1:
            print(f"Key {key} found at position {result}")
        else:
            print(f"key {key} not found")

        time_taken=(end-start)*1000
        print(f"Time taken to search a key element={time_taken}
milliseconds\n")
        results.append((n,time_taken))
    return results

def plot_results(results):
    n_values=[result[0] for result in results]
    times=[result[1] for result in results]

    plt.figure()
    plt.plot(n_values,times,'gD--')
    plt.xlabel("Number of elements(n)")
    plt.ylabel("Time Taken(milliseconds)")
    plt.title("Linear Search Time Complexity")
    plt.grid(True)
    plt.show()

#min
print("Program Name:Write a program to implement linear search
algorithm")

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print("Programmer name: Venu Madhava.M")
r=int(input("Enter the number of inputs:"))
results=linear_key(r)
plot_results(results)
```

Program Name:Write a program to implement linear search algorithm
Programmer name: Venu Madhava.M

Enter the number of inputs: 3
Enter the number of elements: 3

Enter the elements of an array: 1 2 3

Enter the key element to be searched: 3

Key 3 found at position 2
Time taken to search a key element=1.5056133270263672 milliseconds

Enter the number of elements: 5

Enter the elements of an array: 10 20 30 40 50

Enter the key element to be searched: 40

Key 40 found at position 3
Time taken to search a key element=3.8077831268310547 milliseconds

Enter the number of elements: 6

Enter the elements of an array: 20 30 40 10 22 33

Enter the key element to be searched: 22

Key 22 found at position 4
Time taken to search a key element=1.6973018646240234 milliseconds

